SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions (5 Questions × 10 Marks)
Weightage:	35% of CIA	**Total Marks:**	
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	CSA-AI	**Date:**	28/10/2026

Code of Conduct

I B VAMSI/192472327 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VAMSI

QUESTIONS: 5 **Total Marks: 50**

Annexure A Scenario-Based Qns

Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

Introduction

Digital Image Processing (DIP) is a fundamental field that enables computers to enhance, analyse and extract meaningful information from digital images. The five scenario-based questions in this annexure cover core techniques used across medical imaging, surveillance, remote sensing, industrial inspection and agricultural monitoring.

Each question presents a real-world problem, requires selection of an appropriate algorithm, and asks for calculation or qualitative explanation of results. The solutions below provide theoretical background, step-by-step application, numerical examples (where applicable), comparative tables and illustrative diagrams.

All intensity values are assumed to lie in the range [0, L−1] where L = 256 for typical 8-bit images, unless otherwise stated.

Q1. Image Enhancement Using Grey-Level Transformations

Scenario Recap

A hospital's digital X-ray system produces images that appear too dark, making important anatomical structures difficult to identify. Grey-level transformation techniques are to be applied to improve brightness and contrast.

Theory of Grey-Level Transformations

A grey-level (or intensity) transformation maps each input intensity r to an output intensity s according to a function s = T(r). Common families of transformations include:

- Linear transformations: identity, negative image, contrast stretching.
- Logarithmic transformation: $s = c \cdot \log(1 + r)$ – expands dark values, compresses bright values.
- Power-law (gamma) transformation: $s = c \cdot r^{\gamma}$ – for $\gamma < 1$ expands dark regions (brightens the image); for $\gamma > 1$ expands bright regions.

Because X-ray images in this scenario are predominantly dark (most pixel values clustered near the lower end of the intensity scale), a power-law transformation with $\gamma < 1$ or a logarithmic transformation is most suitable.

Chosen Transformation: Power-Law (Gamma Correction)

We select the power-law transformation with c = 1 and $\gamma = 0.4$ (a common clinical choice for dark radiographs):

$$s = r^{0.4}$$

where both r and s are normalised to the range [0, 1]. After transformation the values are scaled back to [0, 255].

Calculation of Transformed Intensity Values

Consider five representative pixels taken from a dark X-ray region. Original intensities (0–255) are first normalised, transformed, then scaled back:

Pixel	Original r	r (norm)	s = r^0.4	s (0–255)	Δ (gain)
P1	20	0.078	0.372	95	+75
P2	35	0.137	0.455	116	+81
P3	55	0.216	0.548	140	+85
P4	80	0.314	0.640	163	+83
P5	110	0.431	0.735	187	+77

Table 1. Sample pixel intensity transformation using s = r^0.4

How the Transformation Improves Visibility

The power-law curve with $\gamma < 1$ is concave-down, meaning it allocates a larger portion of the output dynamic range to the darker input values. Consequently:

- Dark anatomical structures (soft tissue, faint fractures) that occupied only a narrow band of low intensities are stretched across a much wider range of grey levels.
- Local contrast within previously underexposed regions increases, making edges and subtle density differences more perceptible to the radiologist.
- Overall image brightness rises without saturating the already bright areas (bones, metal implants) excessively.

Q2. Histogram Processing and Spatial Filtering

Scenario Recap

A security surveillance system captures night-time images that suffer from poor contrast and unwanted noise. The system must enhance contrast via histogram equalisation and reduce noise with spatial filtering while preserving important details of people and objects.

Histogram Equalisation

Histogram equalisation redistributes intensity values so that the cumulative distribution function (CDF) of the output image becomes approximately linear. The transformation is:

$$s_k = (L-1) \cdot \Sigma\, (n_j / N) \quad \text{for } j = 0 \ldots k$$

where n_j is the number of pixels with intensity j, N is the total number of pixels, and $L = 256$. The result is an image whose histogram is as uniform as possible, thereby maximising global contrast.

Worked Numerical Example (Simplified 3-bit image)

Consider an 8×8 image ($N = 64$) with the following intensity histogram ($L = 8$ levels):

r_k	0	1	2	3	4	5	6/7
n_k	12	18	16	10	5	2	1
CDF	12	30	46	56	61	63	64
s_k	1	3	5	6	7	7	7

Table 2. Histogram equalisation mapping for a simplified 3-bit image

Dark levels (0–2) that occupied most of the histogram are mapped to a wider set of output levels, dramatically improving contrast for night-time scenes.

Spatial Filtering for Noise Reduction

After equalisation, residual sensor noise and quantisation artefacts remain. Two common linear/non-linear spatial filters are considered:

- Averaging (mean) filter – replaces each pixel by the mean of its neighbourhood. Effective against Gaussian noise but blurs edges.
- Median filter – replaces each pixel by the median of its neighbourhood. Excellent for salt-and-pepper noise and preserves sharp edges better than the mean filter.

For surveillance imagery the median filter (typically 3×3 or 5×5) is preferred because it removes impulse noise while keeping the silhouettes of people and objects crisp.

Comparison of the Two Techniques

Aspect	Histogram Equalisation	Median Spatial Filter
Primary goal	Increase global contrast	Suppress noise
Operates on	Entire intensity histogram	Local neighbourhood
Effect on edges	May enhance or create false edges	Preserves edges well
Best used for	Low-contrast night images	Impulse / salt-pepper noise
Typical order	Applied first	Applied after equalisation

Table 3. Comparison of histogram equalisation and median filtering

Together the two techniques produce surveillance frames in which people and objects are both brighter/more contrasted and free of distracting noise, enabling reliable detection and tracking.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

Scenario Recap

Satellite images of land contain high-frequency noise and blurred boundaries between geographical regions. Frequency-domain filtering is required: a low-pass filter to smooth noise and a high-pass filter to sharpen edges.

Frequency Domain Fundamentals

Any digital image f(x,y) can be transformed into its frequency representation F(u,v) by the Discrete Fourier Transform (DFT). Low frequencies correspond to smooth, slowly varying intensity regions; high frequencies correspond to edges, fine detail and noise.

Filtering is performed by multiplying the DFT by a filter transfer function H(u,v) and then computing the inverse DFT:

$$G(u,v) = H(u,v) \cdot F(u,v) \quad \rightarrow \quad g(x,y) = \text{IDFT}\{G\}$$

Low-Pass Filter (Smoothing)

An ideal low-pass filter (ILPF) retains all frequencies inside a radius D0 from the origin and rejects all others:

$$H(u,v) = 1 \text{ if } D(u,v) \leq D0 \; ; \quad 0 \text{ otherwise}$$

Because the ideal filter introduces ringing (Gibbs phenomenon), a Gaussian low-pass filter is often preferred in practice:

$$H(u,v) = \exp\left(-D^2(u,v) / (2\sigma^2) \right)$$

Applying an LPF removes high-frequency sensor noise and mild atmospheric artefacts, producing a smoother image that is easier for subsequent classification algorithms to process.

High-Pass Filter (Sharpening)

A high-pass filter is the complement of a low-pass filter. The ideal high-pass filter is:

$$H(u,v) = 0 \text{ if } D(u,v) \leq D0 \; ; \quad 1 \text{ otherwise}$$

High-pass filtering accentuates edges and fine texture that define boundaries between forests, water bodies, urban areas and agricultural plots, thereby improving visual interpretation and automated edge-based segmentation.

Differences and Contribution to Image Interpretation

Characteristic	Low-Pass Filter	High-Pass Filter
Frequencies kept	Low (smooth areas)	High (edges & noise)
Visual effect	Smoothing / blurring	Sharpening / edge emphasis
Noise behaviour	Reduces noise	Amplifies noise
Typical use in satellite imagery	Pre-processing before classification	Highlighting region boundaries
Risk	Loss of fine detail	Ringing / noise boost

Table 4. Comparison of low-pass and high-pass frequency domain filters

In practice a mild Gaussian LPF is first applied to suppress noise; a subsequent high-pass or unsharp-masking step then restores and enhances the geographical boundaries needed for accurate land-cover interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

Scenario Recap

An automated manufacturing inspection system must detect cracks and surface defects on metal components. Images exhibit brightness variations that complicate direct defect detection. Edge detection followed by thresholding is required to produce a binary defect map.

Edge Detection

Edges are locations of abrupt intensity change. Gradient-based operators approximate the first derivative:

$$\nabla f = [\partial f/\partial x \, , \, \partial f/\partial y \,] \quad , \quad \text{magnitude } M = \sqrt{(Gx^2 + Gy^2)}$$

Popular discrete approximations include the Sobel, Prewitt and Roberts operators. The Sobel operator uses the 3×3 kernels:

Gx (horizontal)	Gy (vertical)
−1 0 +1 −2 0 +2 −1 0 +1	−1 −2 −1 0 0 0 +1 +2 +1

Table 5. Sobel operator kernels

After convolution the gradient magnitude image highlights crack boundaries regardless of the absolute brightness of the surrounding metal surface, thereby overcoming the illumination variation problem.

Thresholding to Obtain Binary Defect Image

A global or adaptive threshold T is applied to the gradient magnitude:

$$g(x,y) = 1 \text{ if } M(x,y) \geq T \; ; \quad 0 \text{ otherwise}$$

The resulting binary image contains white pixels only where strong edges (candidate cracks) exist. Subsequent morphological cleaning (opening/closing) can remove isolated noise pixels and fill small gaps in crack contours.

How the Techniques Help the Inspection System

- Edge detection is largely invariant to slow illumination changes, solving the brightness-variation problem stated in the scenario.
- Thresholding converts the continuous gradient map into a compact binary representation that is easy for a decision module to analyse (e.g., count connected components, measure crack length).
- The combination yields a robust, real-time defect mask suitable for automated accept/reject decisions on the production line.

Q5. Region-Based Segmentation and Boundary Detection

Scenario Recap

An agricultural monitoring system analyses aerial images containing crop fields, water bodies and surrounding land. Regions have similar intensities and some boundaries are discontinuous. Region-based segmentation together with edge linking is required to delineate the different land-cover classes accurately.

Region-Based Segmentation

Region-based methods group neighbouring pixels that share similar characteristics (intensity, colour, texture). Two classical approaches are:

- Region growing – start from seed points and iteratively add adjacent pixels whose intensity difference is below a threshold.
- Split-and-merge (quadtree) – recursively split non-homogeneous regions and later merge adjacent homogeneous ones.

A simple intensity-based region-growing criterion is:

$$|I(x,y) - \mu_region| \leq T$$

where μ_region is the current mean intensity of the growing region and T is a similarity threshold. This process naturally produces closed, labelled regions corresponding to individual crop fields or water bodies even when edges are partially missing.

Edge Linking and Boundary Detection

Edge detectors frequently produce broken contours because of noise or gradual intensity transitions. Edge-linking techniques reconnect these fragments:

- Local processing – examine a small neighbourhood around each edge pixel and link pixels whose gradient magnitude and orientation are similar.
- Global processing (Hough transform) – detect parametric curves (straight field boundaries, circular irrigation ponds) even when many edge pixels are missing.
- Graph-based methods – treat edge pixels as nodes and find the lowest-cost path that closes a contour.

Once continuous boundaries are obtained they can be used either to refine the region-growing result or as an independent segmentation (edge-based segmentation).

Benefit for Agricultural Monitoring

- Region growing groups pixels of similar spectral response into coherent field objects, enabling accurate area measurement and crop-type classification.
- Edge linking restores discontinuous boundaries caused by shadows, clouds or gradual transitions, improving the geometric fidelity of field outlines.
- The combined pipeline yields a clean land-cover map that supports yield estimation, irrigation planning and change detection over successive seasons.

Summary of Techniques Applied

Question	Domain / Method	Key Algorithm	Primary Benefit
Q1	Spatial domain	Power-law γ = 0.4	Brightens dark X-rays
Q2	Histogram + Spatial	Equalisation + Median	Contrast + noise removal
Q3	Frequency domain	LPF + HPF	Noise ↓ , edges ↑
Q4	Edge + Threshold	Sobel + Binary T	Crack detection map
Q5	Region + Boundary	Region growing + linking	Accurate land parcels

Table 6. Overview of techniques used across all five scenarios

Concluding Remarks

The five scenarios illustrate the complementary nature of spatial-domain, frequency-domain, edge-based and region-based methods. In real systems these techniques are frequently combined into processing pipelines: intensity transformation and histogram equalisation for pre-enhancement, spatial or frequency filtering for noise control, and finally edge or region segmentation for high-level interpretation.

Mastery of the underlying mathematics (transformation functions, DFT, gradient operators, region similarity criteria) together with an understanding of their practical trade-offs enables the design of robust image-processing solutions for medicine, security, remote sensing, manufacturing and agriculture.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____